

Modeling and Forecasting Monthly WTI Crude Oil Prices Using ARIMA and ARDL: The Role of the U.S. Dollar Index and the Federal Funds Rate, 2010–2025

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Abstract

This study models and forecasts monthly West Texas Intermediate (WTI) crude oil prices from 2010 to 2025 using both univariate and multivariate time-series approaches. The first objective is to evaluate whether past WTI price dynamics can provide useful short-run forecasts through an Autoregressive Integrated Moving Average (ARIMA) model. The second objective is to examine whether the U.S. Dollar Index and the Federal Funds Effective Rate help explain the short-run and long-run movements of WTI prices using an Autoregressive Distributed Lag (ARDL) framework. Daily WTI and U.S. Dollar Index data are aggregated to monthly frequency and matched with monthly interest rate data, producing a clean balanced sample of 192 monthly observations with no missing values. Exploratory data analysis shows strong regime shifts in WTI prices, especially during the 2014–2016 oil price collapse, the COVID-19 shock in 2020, and the 2021–2022 energy market recovery. The preliminary evidence also suggests a negative relationship between WTI prices and the U.S. Dollar Index, while the role of interest rates appears more dynamic and cannot be inferred from simple correlations alone. These findings motivate the use of ARIMA as a univariate forecasting benchmark and ARDL as a macro-financial relationship model.

Keywords: WTI crude oil price; ARIMA; ARDL; U.S. Dollar Index; Federal Funds Rate; forecasting; time series

JEL Classification: C22; C32; E43; E44; Q43

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1 Introduction

Crude oil is one of the most important commodities in the global economy. Its price affects production costs, inflation, transportation, energy consumption, business investment, and macroeconomic stability. WTI crude oil is especially relevant because it is one of the major benchmark prices for North American oil markets. Previous studies emphasize that crude oil prices are difficult to forecast because they are affected by supply and demand conditions, geopolitical events, financial market expectations, exchange rate movements, and macroeconomic policy conditions (Rodhan and Jaaz, 2022; Hidayat and Wiharja, 2024).

The period from 2010 to 2025 provides a useful setting for examining WTI price dynamics because it includes several major oil market regimes. The early 2010s were characterized by relatively high oil prices, followed by the sharp oil price collapse during 2014–2016. The sample also includes the COVID-19 shock in 2020, the strong post-pandemic recovery in 2021–2022, and the high-interest-rate environment after 2022. These features make the period suitable for studying both forecasting performance and macro-financial relationships.

This study focuses on two explanatory macro-financial variables: the U.S. Dollar Index and the Federal Funds Effective Rate. The U.S. Dollar Index is included because crude oil is priced internationally in U.S. dollars. When the dollar strengthens, crude oil becomes more expensive for non-dollar buyers, which may reduce demand and put downward pressure on oil prices. This expected negative relationship between the U.S. dollar and crude oil prices is commonly discussed in the oil price and exchange rate literature (Siddiqui et al., 2023; Rodhan and Jaaz, 2022). The Federal Funds Effective Rate is included because monetary policy can affect oil prices through financial conditions, demand expectations, investment, inflation, and the exchange rate channel. Recent macroeconomic research also emphasizes the connection between oil price shocks, monetary policy, and inflation dynamics (Gagliardone and Gertler, 2023).

The research questions of this study are as follows:

1. Can ARIMA effectively model and forecast monthly WTI crude oil prices during 2010–2025?
2. What are the short-run and long-run relationships between WTI crude oil prices, the U.S. Dollar Index, and the Federal Funds Effective Rate?
3. Do macro-financial variables such as the U.S. Dollar Index and interest rates improve the understanding of WTI crude oil price dynamics beyond a purely univariate ARIMA model?

Based on these questions, the study tests four main hypotheses. First, monthly WTI

prices are expected to be non-stationary in levels but stationary after differencing. Second, the U.S. Dollar Index is expected to have a negative relationship with WTI prices. Third, the effect of interest rates on WTI prices is expected to be statistically relevant but potentially ambiguous in sign because interest rates operate through multiple channels. Fourth, if a long-run relationship exists among the variables, deviations from the long-run equilibrium should be corrected through a short-run adjustment mechanism.

The contribution of this study is twofold. First, it provides a monthly forecasting exercise for WTI prices using ARIMA and compares fixed multi-step forecasts with rolling one-step-ahead forecasts. Second, it extends the analysis beyond univariate forecasting by using ARDL to examine the role of the U.S. Dollar Index and the Federal Funds Rate in explaining WTI price movements. In this way, the paper combines forecasting evaluation with economic interpretation.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data and variable construction. Section 4 presents the exploratory data analysis. The next sections will discuss stationarity testing, ARIMA modeling, ARDL estimation, forecast evaluation, and conclusions.

2 Literature Review

2.1 WTI crude oil price forecasting and ARIMA models

A large part of the time-series forecasting literature uses ARIMA models to study crude oil prices. ARIMA is useful because it models a variable using its own past values and past forecast errors after the series has been transformed into a stationary process. In the Box–Jenkins framework, the usual steps include checking stationarity, identifying possible AR and MA orders from ACF and PACF plots, estimating candidate models, and checking residual diagnostics before producing forecasts ([Hidayat and Wiharja, 2024](#)).

Several previous studies apply ARIMA directly to WTI crude oil prices. [Rodhan and Jaaz \(2022\)](#) use WTI data from March 1990 to March 2021 and emphasize that stationarity and autocorrelation analysis are necessary before selecting an ARIMA model. Their study finds that WTI prices are difficult to forecast because oil prices fluctuate sharply across different historical episodes. Similarly, [Hidayat and Wiharja \(2024\)](#) apply ARIMA to monthly WTI prices and show that ARIMA can provide useful short-run forecasts, although the selected order can differ depending on sample period and data frequency. These studies support the use of ARIMA in this paper as a univariate benchmark model for WTI price forecasting.

However, ARIMA has an important limitation. Since it uses only the past values of WTI prices, it does not explicitly include macro-financial variables such as exchange rates or interest rates. Therefore, ARIMA is useful as a forecasting benchmark, but it cannot

directly explain how the U.S. Dollar Index or monetary policy conditions are related to WTI prices.

2.2 Oil prices, exchange rates, and the U.S. dollar

The relationship between oil prices and exchange rates is widely discussed in the literature. Since crude oil is traded internationally in U.S. dollars, movements in the dollar can affect global oil demand and pricing. A stronger dollar can make oil more expensive in local currency terms for non-U.S. buyers, which may reduce demand and put downward pressure on oil prices. For this reason, a negative relationship between the U.S. dollar and crude oil prices is often expected ([Rodhan and Jaaz, 2022](#)).

A systematic literature review by [Siddiqui et al. \(2023\)](#) shows that the oil price–exchange rate relationship has been studied across many dimensions, including short-run and long-run relationships, time-varying dependence, volatility spillovers, and crisis-period dynamics. This indicates that a simple static correlation is not enough to understand the relationship. Instead, a dynamic time-series model is more appropriate because the relationship may vary across periods and may involve lagged effects. This provides the motivation for including the U.S. Dollar Index in the ARDL model.

2.3 Oil prices, interest rates, and monetary policy

The Federal Funds Effective Rate is included to capture the role of U.S. monetary policy conditions. Interest rates may affect crude oil prices through several channels. Higher interest rates can reduce aggregate demand, increase financing costs, strengthen the U.S. dollar, and influence expectations about future growth. These channels suggest that the relationship between interest rates and oil prices may be dynamic and may not have a simple fixed sign.

Recent macroeconomic research has renewed attention to the interaction between oil prices and monetary policy. [Gagliardone and Gertler \(2023\)](#) emphasize that oil price shocks and monetary policy conditions played an important role in recent inflation dynamics. Although their framework differs from the ARDL approach used in this study, their analysis supports the broader idea that oil prices and monetary policy are connected through macroeconomic adjustment mechanisms. Therefore, including the Federal Funds Effective Rate allows this paper to examine whether monetary policy conditions help explain WTI price movements.

2.4 ARDL models and the research gap

ARDL models are widely used when the objective is to examine both short-run and long-run relationships among time-series variables. A key advantage of the ARDL framework is

that it can be applied when variables are integrated of order zero, $I(0)$, or order one, $I(1)$, as long as none of the variables is integrated of order two, $I(2)$. In the oil price literature, ARDL has been used to study the relationship between crude oil prices, exchange rates, and macroeconomic variables (Musa et al., 2019).

The existing literature provides two motivations for this paper. First, ARIMA studies show that WTI prices can be forecasted using their own past dynamics, but such models do not directly account for macro-financial variables. Second, the oil price–exchange rate and oil price–monetary policy literatures suggest that WTI prices may be related to the U.S. dollar and interest rates through dynamic channels. This study therefore combines ARIMA and ARDL in one empirical framework. ARIMA is used as a univariate forecasting benchmark, while ARDL is used to examine whether the U.S. Dollar Index and the Federal Funds Rate help explain WTI price dynamics.

3 Data and Variable Construction

3.1 Data sources and sample period

This study uses monthly data from January 2010 to December 2025. The final dataset contains 192 monthly observations and three variables: WTI crude oil price, the U.S. Dollar Index, and the Federal Funds Effective Rate. Daily WTI crude oil prices and the U.S. Dollar Index are aggregated to monthly frequency, while the interest rate series is aligned directly at the monthly level.

The dependent variable is the WTI crude oil price, measured in U.S. dollars per barrel. The U.S. Dollar Index is used as the exchange-rate-related explanatory variable. The Federal Funds Effective Rate is used as the interest rate variable. Table 1 summarizes the variables used in this study.

Table 1: Variable description

Variable	Description	Transformation used
WTI_Oil	WTI crude oil price, USD per barrel	Level and log
USD_Index	U.S. Dollar Index	Level and log
Interest_Rate	Federal Funds Effective Rate	Level and first difference

3.2 Data cleaning and transformation

The cleaned dataset is balanced and contains no missing observations. The monthly date index is continuous from January 2010 to December 2025. This is important because ARIMA and ARDL models require a properly ordered time-series structure.

For the modeling stage, WTI prices and the U.S. Dollar Index are transformed into logarithms:

$$\text{Log_WTI}_t = \log(\text{WTI_Oil}_t), \quad (1)$$

$$\text{Log_USD}_t = \log(\text{USD_Index}_t). \quad (2)$$

Monthly log changes are computed as first differences of the log series:

$$\Delta \text{Log_WTI}_t = \text{Log_WTI}_t - \text{Log_WTI}_{t-1}, \quad (3)$$

$$\Delta \text{Log_USD}_t = \text{Log_USD}_t - \text{Log_USD}_{t-1}. \quad (4)$$

The interest rate is kept in level form for ARDL because it is already measured as a percentage rate and contains values close to zero during several periods. For short-run descriptive analysis, the first difference of the interest rate is computed as:

$$\Delta \text{Interest_Rate}_t = \text{Interest_Rate}_t - \text{Interest_Rate}_{t-1}. \quad (5)$$

The use of log WTI and log USD allows the coefficients of these variables to be interpreted approximately in percentage terms. The interest rate is not logged because a level specification is easier to interpret and avoids problems associated with near-zero values.

4 Exploratory Data Analysis

The purpose of the exploratory data analysis is to describe the main time-series features of WTI prices, the U.S. Dollar Index, and the Federal Funds Rate before formal modeling. This section focuses on trends, regime changes, volatility, distributional properties, seasonal patterns, and preliminary correlations. The EDA is descriptive and does not provide causal evidence. Formal inference is conducted later using ARIMA and ARDL models.

4.1 Time-series trends

Figure 1 plots the three variables over time. WTI crude oil prices show several distinct regimes. During 2010–2013, WTI prices remained relatively high. From 2014 to 2016, prices declined sharply, reflecting the global oil price collapse. After a moderate recovery during 2017–2019, WTI experienced a severe decline during the COVID-19 shock in 2020.

Prices then recovered strongly during 2021–2022 before gradually declining during the later part of the sample.

The U.S. Dollar Index shows a more gradual upward movement over the full sample. It increased strongly during the mid-2010s and again around 2022. The Federal Funds Rate displays clear monetary policy regimes. It remained near zero in the early part of the sample, increased during the 2016–2019 tightening cycle, fell sharply again during the COVID-19 period, and rose rapidly during the 2022–2023 tightening cycle.

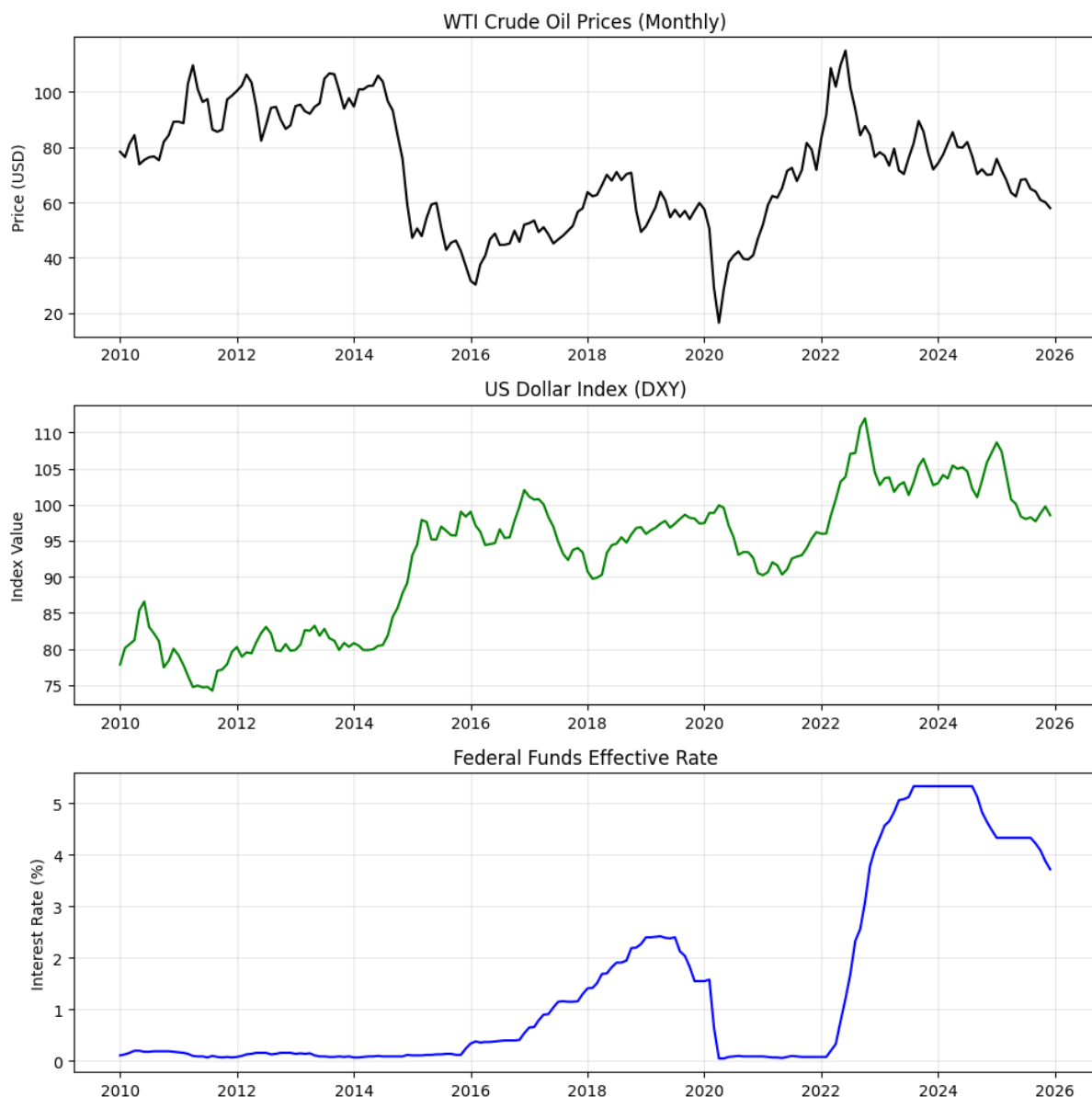


Figure 1: Monthly WTI crude oil price, U.S. Dollar Index, and Federal Funds Rate

The time-series plots suggest that the variables are likely non-stationary in levels. WTI prices in particular display strong regime shifts and irregular shocks. This motivates the later use of formal unit root tests before ARIMA and ARDL estimation.

4.2 Year-over-year changes

Figure 2 shows the year-over-year changes in the variables. The year-over-year change in WTI prices highlights the extreme size of the COVID-19 shock and the subsequent rebound. The large positive year-over-year growth after 2020 should be interpreted carefully because it partly reflects a base effect from the very low oil price level during the COVID-19 collapse.

The U.S. Dollar Index shows strong appreciation episodes around the mid-2010s and 2022. The interest rate series captures the sharp monetary easing during 2020 and the aggressive tightening cycle after 2022. These patterns indicate that the sample contains several macro-financial regimes, which supports the use of dynamic time-series models.

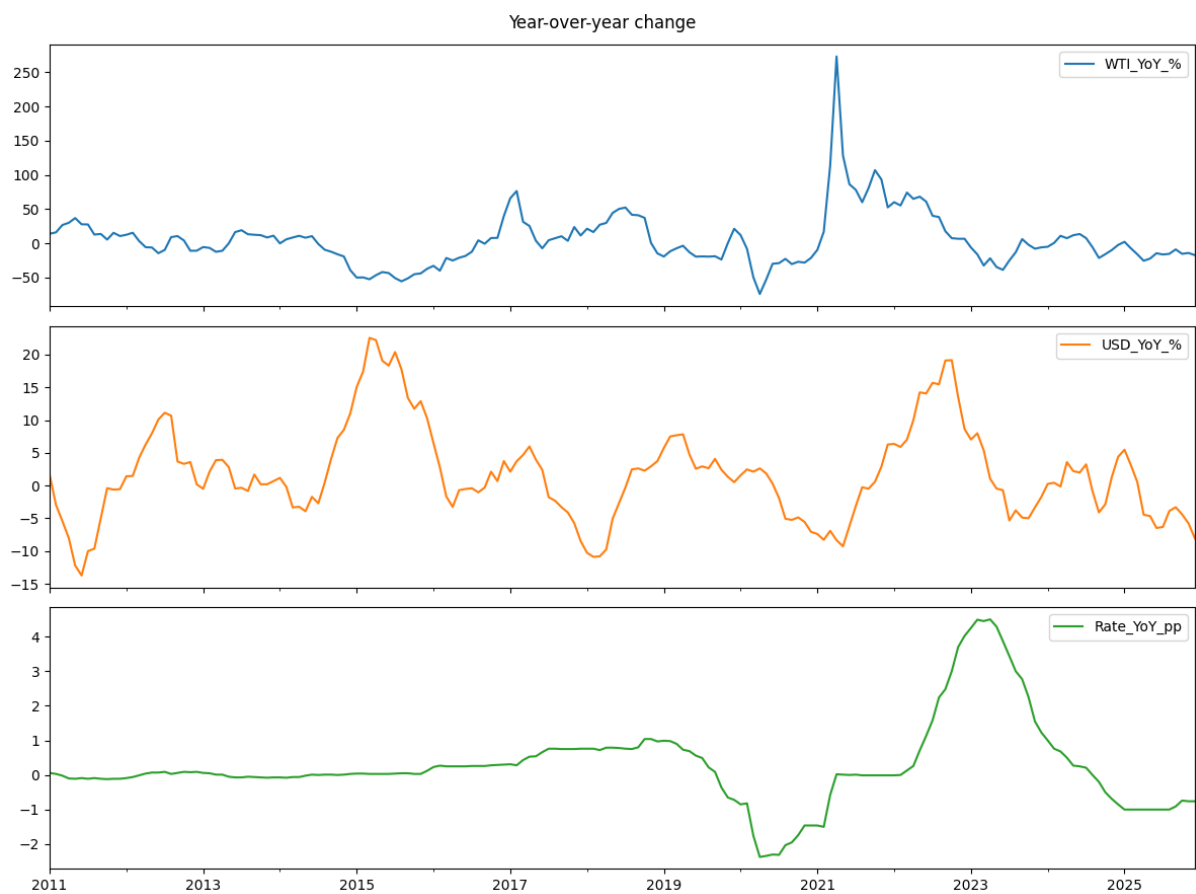


Figure 2: Year-over-year changes in WTI price, U.S. Dollar Index, and Federal Funds Rate

4.3 Normalized comparison of WTI and the U.S. Dollar Index

Because WTI prices and the U.S. Dollar Index are measured in different units, Figure 3 normalizes both series to the first month of the sample. The normalized plot shows that WTI prices are much more volatile than the U.S. Dollar Index. During 2014–2016, WTI prices fell sharply while the U.S. Dollar Index increased, which is consistent with

the expected negative relationship between oil prices and dollar strength.

However, the relationship is not perfectly stable across the full sample. During 2021–2022, both WTI prices and the U.S. Dollar Index increased for part of the period. This suggests that the oil–dollar relationship may vary across regimes and should not be interpreted using simple visual inspection alone. This finding supports the use of the ARDL framework to examine dynamic relationships.

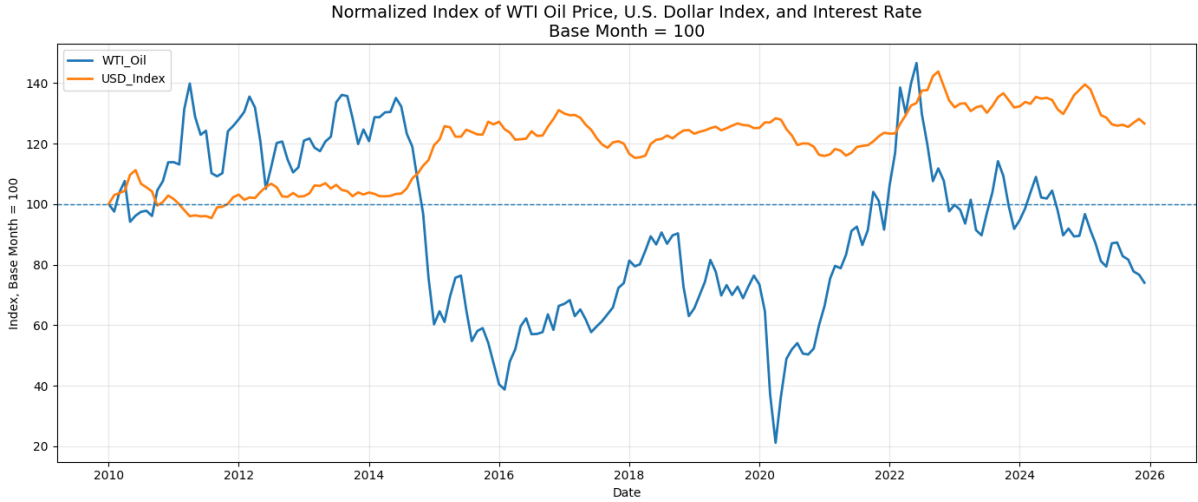


Figure 3: Normalized WTI crude oil price and U.S. Dollar Index

4.4 STL decomposition of log WTI prices

To examine whether WTI prices display stable seasonality, this study applies STL decomposition to the log WTI series using a 12-month seasonal period. Figure 4 separates log WTI prices into trend, seasonal, and residual components.

The trend component captures the major oil market regimes in the sample, including the 2014–2016 collapse, the COVID-19 shock in 2020, and the post-pandemic recovery in 2021–2022. The seasonal component fluctuates around zero and appears relatively small compared with the trend and residual components. This suggests that monthly WTI price movements are driven more by regime shifts and irregular shocks than by stable deterministic seasonality.

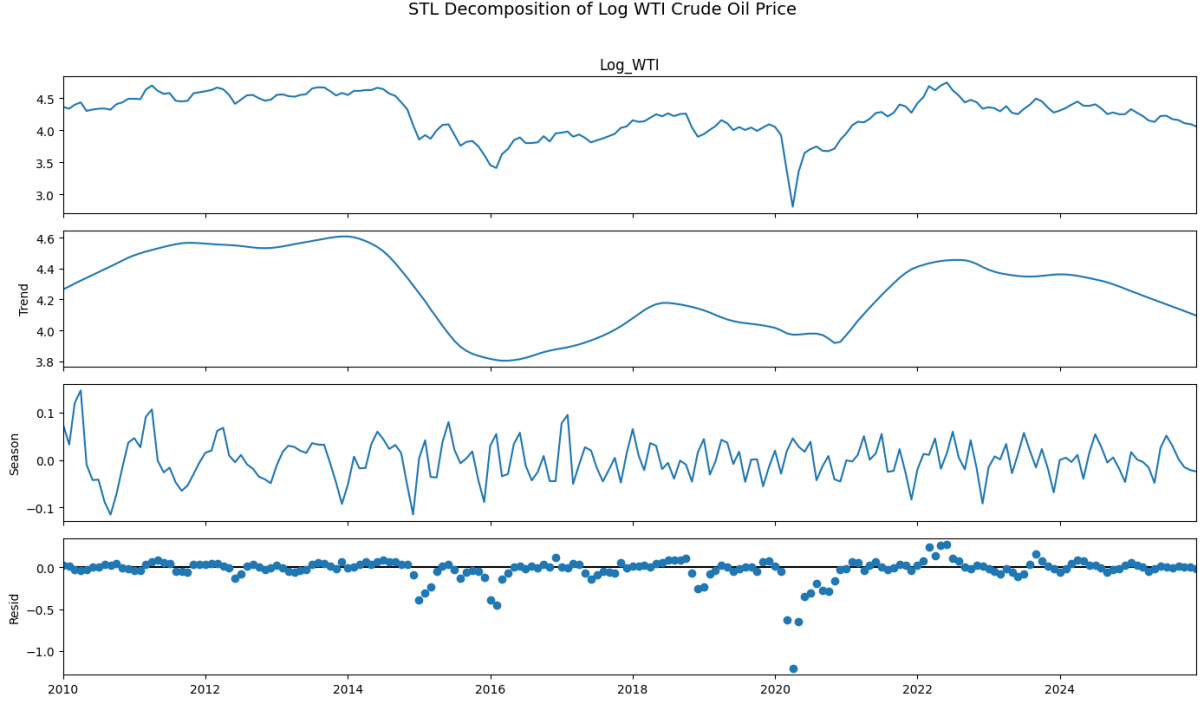


Figure 4: STL decomposition of log WTI crude oil price

This result supports the use of a non-seasonal ARIMA model as a reasonable baseline. Seasonal extensions such as SARIMA may still be considered in future research, but the EDA does not indicate strong and stable seasonality in the WTI price series.

4.5 Distribution of level variables

Figure 5 presents histograms and boxplots of the level variables. WTI prices are widely dispersed, reflecting the high volatility of oil markets. The lower tail is associated with the COVID-19 oil price collapse, while the upper values correspond to high-price periods such as the early 2010s and the 2021–2022 recovery. These extreme observations are retained because they represent genuine market events rather than data errors.

The U.S. Dollar Index shows a clustered distribution, indicating different exchange-rate regimes over the sample period. The index was relatively low in the early 2010s and shifted to higher levels after the mid-2010s. The interest rate distribution is highly right-skewed, with many observations close to zero and fewer observations at high interest rate levels. This reflects distinct monetary policy regimes, including the near-zero interest rate period and the post-2022 tightening cycle.

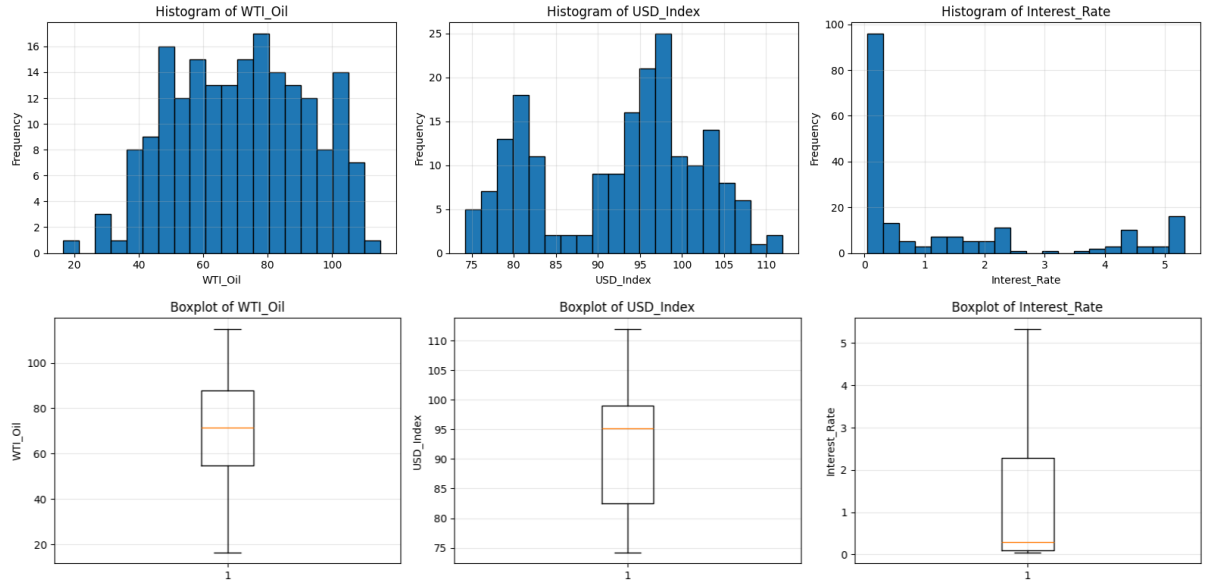


Figure 5: Histograms and boxplots of level variables

4.6 Distribution of transformed variables

Figure 6 presents the distribution of transformed variables: monthly log changes in WTI prices, monthly log changes in the U.S. Dollar Index, and first differences of the Federal Funds Rate. The monthly log changes in WTI prices are centered around zero but show fat tails and extreme observations. This indicates that WTI prices experience large short-run shocks, especially during crisis periods.

Monthly log changes in the U.S. Dollar Index are less volatile than WTI price changes. Changes in the Federal Funds Rate are concentrated around zero, reflecting periods of stable monetary policy, but several large positive and negative changes are observed during tightening and easing episodes.

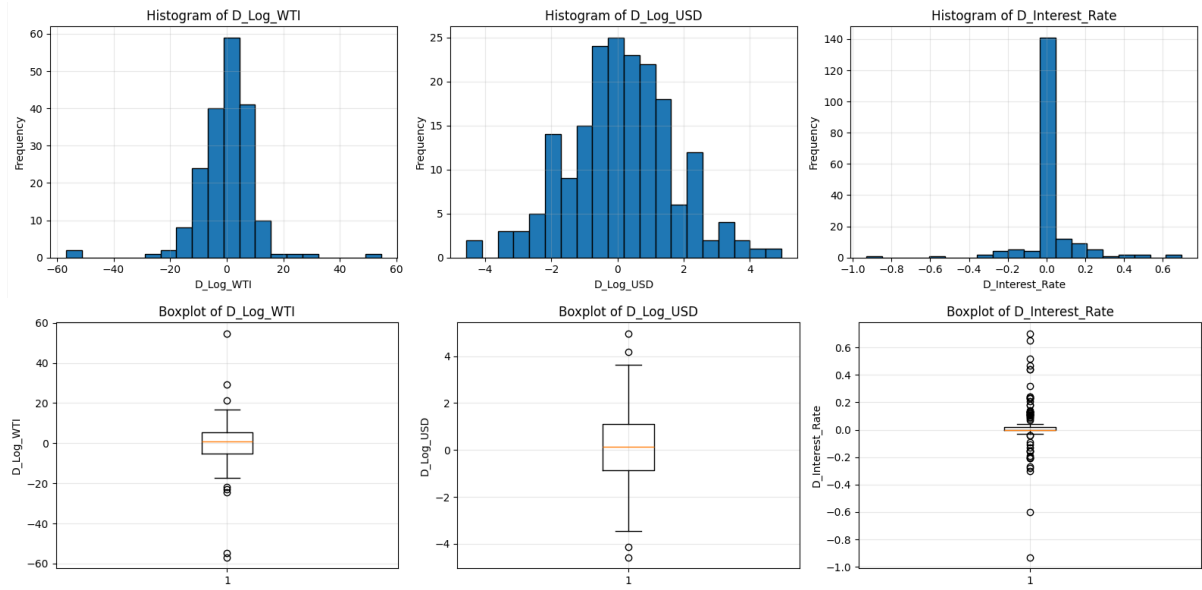


Figure 6: Histograms and boxplots of transformed variables

The distributional evidence suggests that WTI price changes are more volatile and more exposed to extreme shocks than the U.S. Dollar Index and interest rate changes. This is consistent with the broader crude oil volatility literature, which emphasizes that oil prices often display large shocks, fat tails, and time-varying volatility (Ng'ang'a and Oleche, 2022).

4.7 Rolling volatility of WTI price changes

Figure 7 shows the 12-month rolling standard deviation of monthly log changes in WTI prices. The volatility of WTI price changes is not constant over time. Volatility increased during the 2014–2016 oil price collapse, reached its highest level during the COVID-19 shock in 2020, and rose again around the 2022 energy market shock.

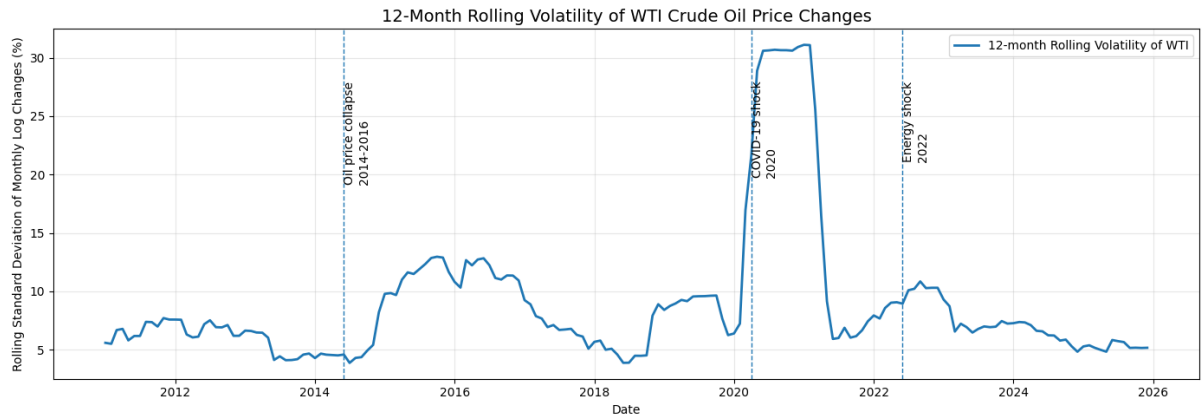


Figure 7: 12-month rolling volatility of WTI crude oil price changes

This confirms that WTI prices are subject to time-varying shocks. Although this study

focuses on ARIMA and ARDL, the volatility pattern suggests that volatility models such as GARCH could be considered as an extension in future research.

4.8 Correlation analysis

Table 2 reports selected correlations among the level and transformed variables. In levels, WTI prices and the U.S. Dollar Index have a moderate negative correlation of approximately -0.47 . This is consistent with the expected inverse relationship between oil prices and dollar strength. The correlation between WTI prices and the interest rate is close to zero, suggesting that a simple contemporaneous level correlation does not capture the role of monetary policy.

After transforming the variables, the correlation between monthly log changes in WTI and monthly log changes in the U.S. Dollar Index becomes weaker, approximately -0.18 . The correlation between WTI log changes and interest rate changes is positive but modest. These differences show that level correlations may reflect common trends or regime shifts and should be interpreted cautiously.

Table 2: Selected correlations among level and transformed variables

Variable pair	Correlation	Interpretation
WTI_Oil and USD_Index	-0.47	Moderate negative correlation in levels
WTI_Oil and Interest_Rate	-0.03	Almost no linear level correlation
USD_Index and Interest_Rate	0.68	Positive correlation in levels
$\Delta\text{Log_WTI}$ and $\Delta\text{Log_USD}$	-0.18	Weak negative short-run correlation
$\Delta\text{Log_WTI}$ and $\Delta\text{Interest_Rate}$	0.26	Modest positive short-run correlation

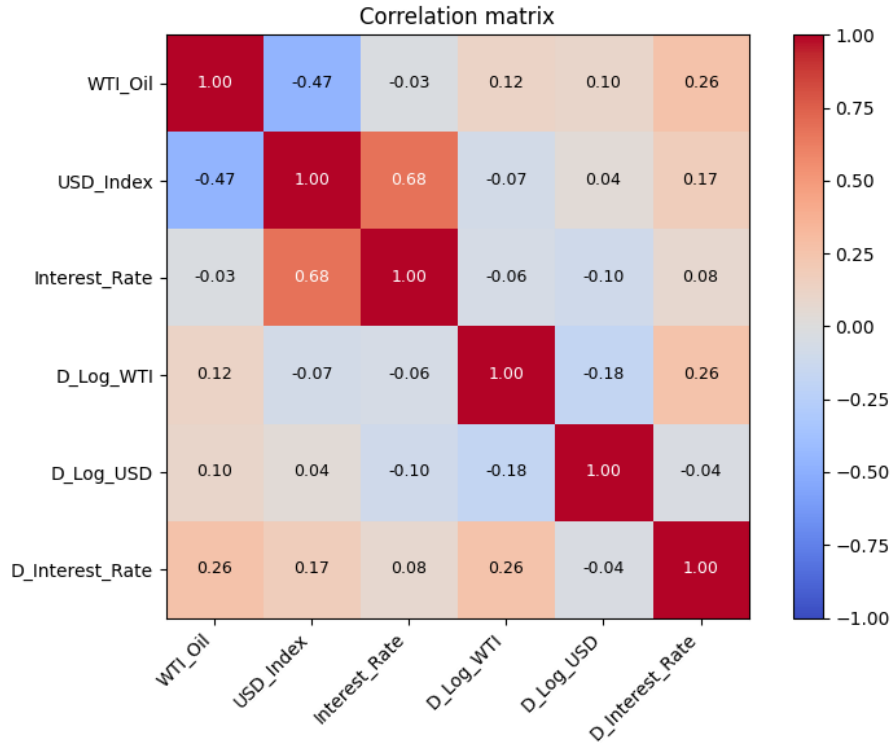


Figure 8: Correlation matrix of level and transformed variables

The correlation analysis provides preliminary descriptive evidence only. Since the variables are likely non-stationary in levels, correlation coefficients should not be treated as formal evidence of economic relationships. The ARDL model is therefore used later to examine the dynamic relationship between WTI prices, the U.S. Dollar Index, and the Federal Funds Rate.

4.9 Summary of EDA findings

The exploratory data analysis provides several important findings. First, WTI crude oil prices display strong regime changes and large shocks over the 2010–2025 period. Second, the U.S. Dollar Index shows an overall upward shift, especially after the mid-2010s and around 2022. Third, the Federal Funds Rate follows distinct monetary policy regimes, including near-zero interest rates and rapid tightening after 2022. Fourth, STL decomposition suggests that WTI prices are driven mainly by trend shifts and irregular shocks rather than stable seasonality. Fifth, rolling volatility shows that WTI price changes are highly time-varying, especially during 2014–2016, 2020, and 2022. Finally, the correlation analysis suggests a negative relationship between WTI prices and the U.S. Dollar Index, while the role of interest rates appears more complex.

These findings motivate the next stage of analysis. Since WTI prices and the explanatory variables appear to contain strong persistence and possible non-stationarity, formal unit root tests are required before model estimation. ARIMA is used as a univariate

forecasting benchmark for WTI prices, while ARDL is used to examine the short-run and long-run roles of the U.S. Dollar Index and the Federal Funds Effective Rate.

5 Model Preparation and Stationarity Tests

The exploratory analysis suggests that the variables contain strong persistence, regime shifts, and possible non-stationarity. Before estimating ARIMA and ARDL models, this section prepares the modeling dataset and conducts unit root tests. This step is important because ARIMA requires the dependent variable to be stationary after differencing, while ARDL bounds testing requires that none of the variables is integrated of order two, $I(2)$.

5.1 Train-test split

For forecasting evaluation, the sample is divided chronologically into a training set and a test set. The training set covers January 2010 to December 2022, while the test set covers January 2023 to December 2025. This split preserves the time ordering of the data and avoids look-ahead bias.

Table 3: Train-test split

Sample	Period	Number of observations	Purpose
Training set	2010-01 to 2022-12	156	Model estimation and selection
Test set	2023-01 to 2025-12	36	Out-of-sample forecast evaluation
Full sample	2010-01 to 2025-12	192	Descriptive analysis

For ARIMA, the dependent variable is the logarithm of WTI prices, denoted by $\log(WTI_t)$. For ARDL, the dependent variable is also $\log(WTI_t)$, while the explanatory variables are $\log(USD_t)$ and the Federal Funds Effective Rate, denoted by R_t .

5.2 Unit root tests

The Phillips–Perron unit root test is applied to the level variables and their first differences. The null hypothesis of the Phillips–Perron test is that the series contains a unit root. Table 4 reports the results for the training sample.

Table 4: Phillips–Perron unit root test results on the training sample

Variable	PP statistic	p-value	Conclusion at 5% level
$\log(WTI_t)$	-2.1301	0.2326	Non-stationary
$\log(USD_t)$	-1.2843	0.6363	Non-stationary
R_t	-0.9362	0.7758	Non-stationary
$\Delta \log(WTI_t)$	-9.1294	0.0000	Stationary
$\Delta \log(USD_t)$	-8.8266	0.0000	Stationary
ΔR_t	-6.1955	0.0000	Stationary

The results indicate that all three level variables are non-stationary, while their first differences are stationary. Therefore, the variables are treated as integrated of order one, $I(1)$. For ARIMA modeling, this supports the use of first differencing for log WTI prices. For ARDL modeling, the result is also appropriate because the ARDL bounds testing framework can be used when variables are $I(0)$ or $I(1)$, provided that no variable is $I(2)$.

6 ARIMA Modeling and Forecasting

6.1 Model specification

The ARIMA model is used as the univariate benchmark for WTI price forecasting. It models WTI prices using only their own past values and past forecast errors. This follows the Box–Jenkins approach, which requires stationarity testing, model identification using autocorrelation patterns, model selection, residual diagnostics, and forecast evaluation (Rodhan and Jaaz, 2022; Hidayat and Wiharja, 2024).

Let $y_t = \log(WTI_t)$. An ARIMA(p, d, q) model can be written as:

$$\Phi_p(L)(1 - L)^d y_t = c + \Theta_q(L)\varepsilon_t, \quad (6)$$

where p is the autoregressive order, d is the differencing order, q is the moving-average order, L is the lag operator, and ε_t is a white-noise error term. Since the unit root tests show that $\log(WTI_t)$ is non-stationary but $\Delta \log(WTI_t)$ is stationary, the differencing order is set to $d = 1$.

6.2 ACF and PACF analysis

The ACF and PACF plots of $\Delta \log(WTI_t)$ are used to guide model identification. The differenced log WTI series shows limited autocorrelation after differencing, with notable spikes at the early lags. This suggests that relatively low-order ARIMA models are appropriate.

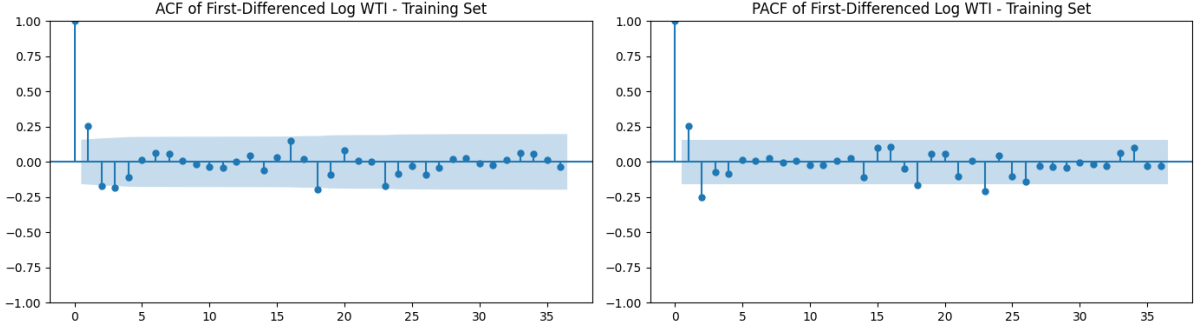


Figure 9: ACF and PACF of first-differenced log WTI prices

6.3 ARIMA model selection

Candidate ARIMA models are estimated on the training sample. Since $d = 1$, the search focuses on combinations of p and q . The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used for in-sample model comparison.

The AIC selects ARIMA(2,1,0), while the BIC selects ARIMA(0,1,1). This difference is expected because BIC imposes a stronger penalty on model complexity. In addition to AIC and BIC, forecast performance is evaluated on the test set using both fixed multi-step forecasts and rolling one-step-ahead forecasts.

Table 5: ARIMA model selection summary

Criterion	Selected model	Interpretation
AIC	ARIMA(2,1,0)	Best in-sample fit among candidates
BIC	ARIMA(0,1,1)	More parsimonious specification
Rolling forecast accuracy	ARIMA(2,1,1)	Best one-step-ahead forecast performance
Benchmark	ARIMA(0,1,0)	Random walk benchmark

6.4 Residual diagnostics

After estimating the candidate ARIMA models, residual diagnostics are conducted to examine whether the selected model adequately captures the autocorrelation structure of WTI prices. Since ARIMA models with differencing may produce unstable initial residuals due to model initialization, the first few residuals are removed before the final diagnostic analysis. For the ARIMA(2,1,0) model, the initial burn-in length is set equal to $p + d + q = 3$ observations.

$$\text{Burn-in length} = p + d + q = 2 + 1 + 1 = 4. \quad (7)$$

The cleaned residuals fluctuate around zero without a clear trend, suggesting that the model captures the main mean dynamics of the differenced log WTI series. The residual

ACF also shows no substantial remaining autocorrelation, indicating that the selected ARIMA model adequately removes the serial dependence in WTI price changes.

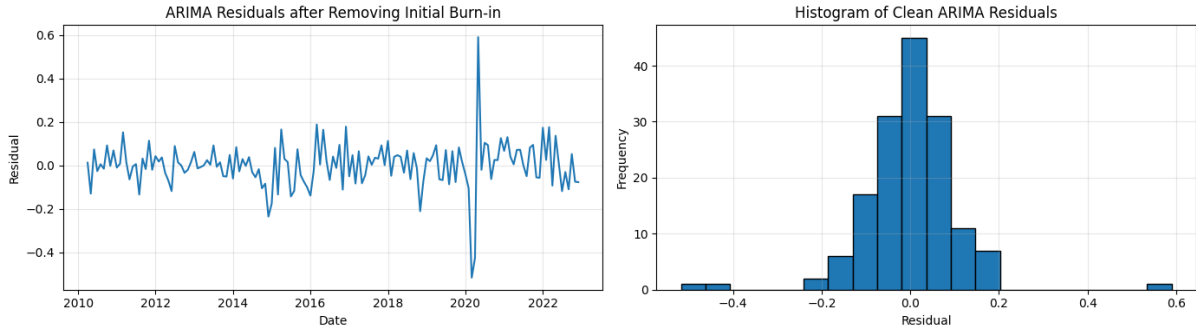


Figure 10: Cleaned residual diagnostics for the selected ARIMA(2,1,0) model

Although the cleaned residuals do not show strong remaining autocorrelation, some large residuals remain around the COVID-19 period. These observations reflect extreme oil market shocks rather than data errors. Therefore, they are retained in the analysis. This result suggests that ARIMA can capture regular short-run time-series dependence, but it remains limited in predicting large exogenous shocks.

The Jarque–Bera test still indicates non-normality in the residuals, which is consistent with the fat-tailed behavior often observed in crude oil price changes. This does not invalidate the ARIMA model as a forecasting benchmark, but it implies that forecast intervals should be interpreted cautiously.

6.5 Fixed multi-step ARIMA forecasts

The fixed multi-step forecast evaluates what the model predicts for 2023–2025 when it is estimated only with information available up to December 2022. The results are shown in Table 6. The best fixed ARIMA forecast is produced by ARIMA(0,1,1), with a MAPE of 8.93%. The random walk benchmark, ARIMA(0,1,0), performs only slightly worse, with a MAPE of 9.57%.

Table 6: Fixed multi-step forecast performance, 2023–2025

Model	MAE	RMSE	MAPE
ARIMA(0,1,1)	6.30	7.63	8.93%
ARIMA(0,1,0)	6.59	8.20	9.57%
ARIMA(2,1,0)	6.70	8.30	9.74%
ARDL(2,1,3), BIC	23.56	25.01	31.50%
ARDL(2,1,4), AIC	26.13	27.41	35.28%

The fixed ARIMA forecasts are relatively flat because ARIMA models with first differencing tend to forecast future log price changes toward zero over longer horizons. This is not necessarily a model error. Instead, it reflects the difficulty of forecasting WTI prices over a long horizon without updating the model with new information.

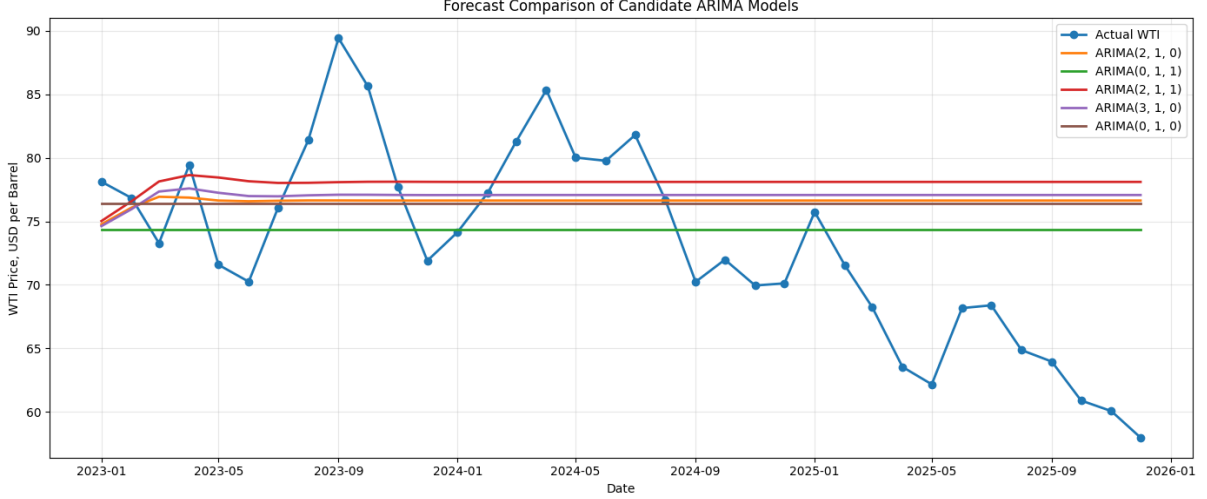


Figure 11: Fixed multi-step ARIMA forecast comparison

6.6 Rolling one-step-ahead ARIMA forecasts

A rolling one-step-ahead forecast is also conducted. In this approach, the model forecasts one month ahead, then the actual observation is added to the information set before forecasting the next month. This procedure avoids look-ahead bias because the model never uses future information when making a forecast. It reflects a realistic monthly forecasting environment in which new information becomes available over time.

The rolling forecast results are reported in Table 7. ARIMA(2,1,1) produces the lowest forecast errors, with a MAPE of 4.82%. However, its improvement over the random walk benchmark is modest. This suggests that monthly WTI prices are highly persistent and that short-run forecasts are strongly driven by the most recent observed price.

Table 7: Rolling one-step-ahead forecast performance, 2023–2025

Model	MAE	RMSE	MAPE
ARIMA(2,1,1)	3.57	4.18	4.82%
ARIMA(3,1,0)	3.62	4.24	4.88%
ARIMA(2,1,0)	3.61	4.26	4.87%
ARIMA(0,1,0)	3.64	4.28	4.91%
ARIMA(0,1,1)	3.68	4.31	4.94%
ARDL(2,1,3), BIC	4.77	6.08	6.43%
ARDL(2,1,4), AIC	4.99	6.25	6.69%

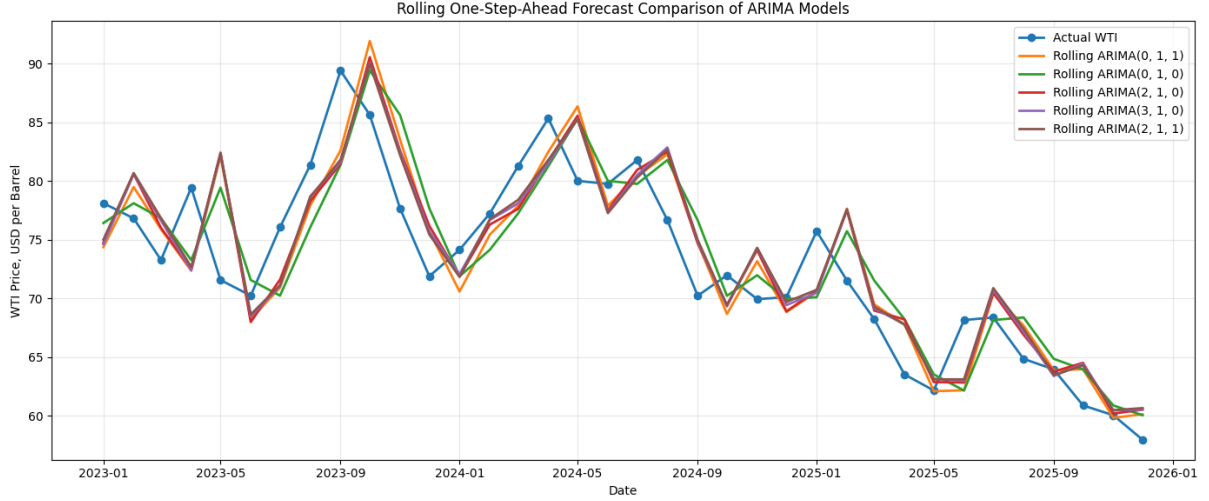


Figure 12: Rolling one-step-ahead ARIMA forecast comparison

Overall, ARIMA performs well for short-run rolling forecasts. However, the small improvement over the random walk benchmark shows that WTI prices are difficult to forecast using past prices alone. This provides motivation for examining whether macro-financial variables such as the U.S. Dollar Index and the Federal Funds Rate help explain WTI price dynamics.

7 ARDL Modeling

7.1 Model specification

The ARDL model is used to examine the dynamic relationship between WTI prices, the U.S. Dollar Index, and the Federal Funds Rate. Unlike ARIMA, ARDL includes explanatory variables and can distinguish between short-run dynamics and possible long-run relationships. This is suitable for the research objective because the oil price–exchange rate literature emphasizes that the relationship may involve short-run, long-run, and time-varying components (Siddiqui et al., 2023). ARDL has also been used in studies of crude oil prices, exchange rates, and macroeconomic variables (Musa et al., 2019).

The baseline ARDL model is specified as:

$$\begin{aligned}
 \log(WTI_t) = & \alpha_0 + \sum_{i=1}^p \phi_i \log(WTI_{t-i}) \\
 & + \sum_{j=0}^{q_1} \beta_j \log(USD_{t-j}) \\
 & + \sum_{k=0}^{q_2} \gamma_k R_{t-k} + \varepsilon_t,
 \end{aligned} \tag{8}$$

where WTI_t is the monthly WTI crude oil price, USD_t is the U.S. Dollar Index, R_t is the Federal Funds Effective Rate, and ε_t is the error term.

7.2 Lag selection

ARDL lag orders are selected using both AIC and BIC. Because the sample size is moderate, BIC is preferred for the main specification since it imposes a stronger penalty on model complexity and reduces the risk of overfitting. The AIC-selected model is ARDL(2,1,4), while the BIC-selected model is ARDL(2,1,3).

Table 8: ARDL lag selection

Criterion	Selected model	Description	Role in this study
AIC	ARDL(2,1,4)	More parameterized model	Robustness check
BIC	ARDL(2,1,3)	More parsimonious model	Preferred specification

The BIC-selected ARDL(2,1,3) model is preferred for three reasons. First, it is more parsimonious than the AIC-selected model. Second, it passes the residual diagnostic tests. Third, it performs slightly better than the AIC model in both fixed dynamic and rolling one-step conditional forecasts. Therefore, ARDL(2,1,3) is used as the main ARDL model.

7.3 ARDL residual diagnostics

The residual diagnostics for the BIC-selected ARDL model are reported in Table 9. The Ljung–Box test p-values are all above 0.05, indicating no evidence of residual autocorrelation. The Jarque–Bera test does not reject normality, and the ARCH test does not detect remaining conditional heteroskedasticity.

Table 9: Residual diagnostics for BIC-selected ARDL(2,1,3)

Diagnostic test	Test statistic / lag	p-value
Ljung–Box	Lag 6	0.8296
Ljung–Box	Lag 12	0.8501
Ljung–Box	Lag 18	0.9027
Ljung–Box	Lag 24	0.7771
Jarque–Bera normality test	–	0.3630
ARCH test	12 lags	0.2997

These results suggest that the BIC-selected ARDL model is well specified in terms of

residual behavior. This supports its use for economic interpretation of the relationship between WTI prices, the U.S. Dollar Index, and the Federal Funds Rate.

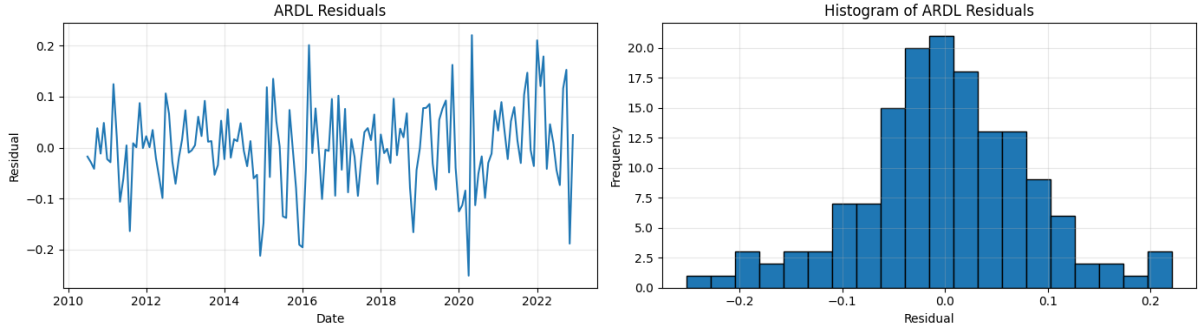


Figure 13: Residual diagnostics for BIC-selected ARDL(2,1,3)

7.4 UECM representation and bounds test

The ARDL model can be rewritten as an unrestricted error correction model (UECM) to test for the existence of a long-run relationship. The bounds test examines the null hypothesis of no cointegration among the variables.

The estimated bounds test statistic is 3.9024. The lower-bound p-value is 0.0185, while the upper-bound p-value is 0.0834. Therefore, the result provides weak evidence of cointegration at the 10% significance level, but the evidence is not strong enough at the 5% level.

Table 10: ARDL bounds test result

Item	Value
Bounds test statistic	3.9024
Lower-bound p-value	0.0185
Upper-bound p-value	0.0834
Conclusion	Weak evidence of cointegration at 10% level

The weak bounds test result means that the long-run coefficients should be interpreted cautiously. The evidence suggests a possible long-run relationship among WTI prices, the U.S. Dollar Index, and the Federal Funds Rate, but it is not strong enough to make a definitive conclusion at the 5% level.

7.5 Error correction and long-run coefficients

In the UECM representation, the coefficient on the lagged level of log WTI is the error correction speed. The estimated coefficient is approximately -0.1197 and is statistically

significant. This indicates that about 12% of deviations from the long-run equilibrium are corrected each month.

$$\text{Speed of adjustment} \approx -0.1197. \quad (9)$$

The long-run coefficients are calculated from the UECM level terms using:

$$\theta_x = -\frac{\delta_x}{\lambda_y}, \quad (10)$$

where δ_x is the coefficient of the lagged level of explanatory variable x , and λ_y is the coefficient of the lagged level of $\log(WTI_t)$.

Table 11: Long-run coefficients from BIC-selected ARDL(2,1,3)

Variable	Long-run coefficient	Interpretation
$\log(USD_t)$	-3.30	Negative long-run association with WTI prices
R_t	0.049	Small positive long-run association

The negative long-run coefficient on the U.S. Dollar Index is consistent with the expected inverse relationship between dollar strength and crude oil prices. A 1% increase in the U.S. Dollar Index is associated with an approximate 3.30% decline in WTI prices in the long run, conditional on the model specification. However, because the bounds test provides only weak evidence of cointegration, this coefficient should be interpreted cautiously.

The long-run coefficient on the Federal Funds Rate is positive but small. This suggests that the role of interest rates is more complex and may operate mainly through short-run dynamics, exchange rate channels, and macroeconomic conditions rather than through a simple long-run effect. This is consistent with the idea that oil prices and monetary policy interact through broader macroeconomic adjustment mechanisms ([Gagliardone and Gertler, 2023](#)).

7.6 ARDL forecasting

The fixed dynamic conditional ARDL forecast performs poorly over the 2023–2025 test period. The BIC-selected ARDL(2,1,3) model produces a MAPE of 31.50%, while the AIC-selected ARDL(2,1,4) model produces a MAPE of 35.28%. This weak performance occurs because dynamic multi-step ARDL forecasts can accumulate errors over time and may be pulled strongly toward the estimated long-run equilibrium.

However, the rolling one-step conditional ARDL forecast performs much better. The BIC-selected ARDL(2,1,3) model achieves a MAPE of 6.43%, which is better than the

AIC-selected ARDL(2,1,4) model with a MAPE of 6.69%. This further supports the choice of the BIC model as the preferred ARDL specification.

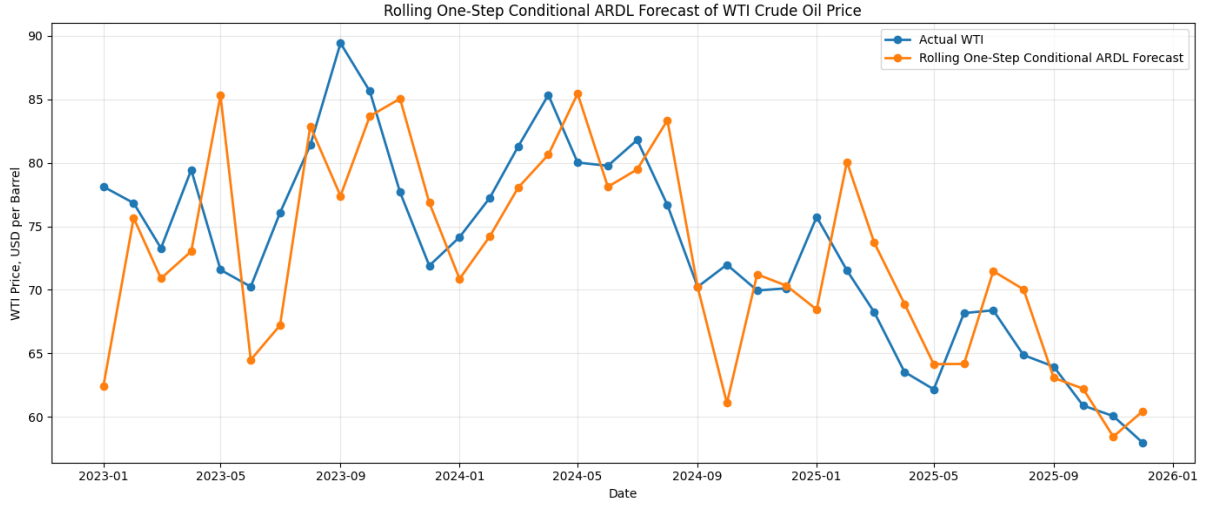


Figure 14: Rolling one-step conditional forecast from BIC-selected ARDL(2,1,3)

The ARDL forecast is conditional because actual out-of-sample values of the U.S. Dollar Index and the Federal Funds Rate are used as explanatory variables. In a fully real-time forecasting exercise, these explanatory variables would also need to be forecasted.

8 Forecast Comparison and Discussion

The forecast results show a clear difference between fixed multi-step forecasts and rolling one-step-ahead forecasts. Fixed forecasts ask what the model predicts for the entire 2023–2025 period using only information available up to December 2022. Rolling forecasts, by contrast, update the information set each month before forecasting the next month.

The ARIMA models perform better than ARDL in fixed multi-step forecasting. ARIMA(0,1,1) has the lowest fixed forecast error among the ARIMA candidates, while fixed ARDL forecasts perform poorly. This indicates that ARDL is not suitable as a long-horizon dynamic forecasting model in this setting.

In rolling one-step-ahead forecasting, ARIMA(2,1,1) performs best, with a MAPE of 4.82%. However, the random walk benchmark ARIMA(0,1,0) also performs well, with a MAPE of 4.91%. The small difference indicates that monthly WTI prices are highly persistent and that simple benchmark models are difficult to outperform.

ARDL performs worse than ARIMA in rolling forecast accuracy, but it remains useful for economic interpretation. The BIC-selected ARDL(2,1,3) model passes residual diagnostic tests and provides evidence of dynamic relationships between WTI prices, the U.S. Dollar Index, and the Federal Funds Rate. The negative long-run coefficient on the U.S.

Dollar Index is consistent with the oil–exchange rate literature, while the interest rate effects appear more dynamic and less straightforward.

The overall interpretation is therefore as follows. ARIMA is better suited for short-run forecasting of WTI prices, especially in a rolling one-step-ahead setting. ARDL is more useful for understanding how macro-financial variables are related to WTI price movements. Thus, the two models serve different but complementary purposes in this study.

9 Limitations

This study has several limitations. First, ARIMA is a univariate model and therefore cannot directly incorporate macroeconomic shocks, geopolitical events, supply disruptions, or monetary policy variables. This explains why fixed multi-step ARIMA forecasts become relatively flat and fail to capture large changes in the test period.

Second, the ARDL model includes only the U.S. Dollar Index and the Federal Funds Rate as explanatory variables. WTI prices are also affected by global oil supply, OPEC decisions, geopolitical tensions, inventories, futures market expectations, and global demand conditions. Excluding these variables may limit the explanatory power of the ARDL model.

Third, the bounds test provides only weak evidence of cointegration at the 10% level and not at the 5% level. Therefore, long-run ARDL coefficients should be interpreted with caution.

Fourth, the EDA shows clear time-varying volatility in WTI price changes. Although this study focuses on ARIMA and ARDL, future research could extend the analysis using GARCH-type models to capture volatility clustering and fat-tailed behavior in oil price changes (Ng’ang’a and Oleche, 2022).

10 Conclusion

This study models and forecasts monthly WTI crude oil prices from 2010 to 2025 using ARIMA and ARDL models. The analysis addresses three main questions: whether ARIMA can forecast monthly WTI prices, whether the U.S. Dollar Index and the Federal Funds Rate help explain WTI price movements, and whether macro-financial variables add useful information beyond a univariate ARIMA model.

The unit root tests show that log WTI prices, log U.S. Dollar Index, and the Federal Funds Rate are non-stationary in levels but stationary after first differencing. This supports the use of ARIMA models with first differencing and validates the use of the ARDL framework since none of the variables appears to be $I(2)$.

The ARIMA results show that fixed multi-step forecasts are relatively flat, reflecting the limitation of univariate ARIMA models in long-horizon forecasting. However, rolling one-step-ahead ARIMA forecasts perform well. ARIMA(2,1,1) achieves the lowest rolling forecast errors, with a MAPE of 4.82%. The improvement over the random walk benchmark is modest, indicating that monthly WTI prices are highly persistent.

The ARDL results show that the BIC-selected ARDL(2,1,3) model is preferred because it is parsimonious, passes residual diagnostics, and performs better than the AIC-selected model in forecast checks. The bounds test provides weak evidence of cointegration at the 10% level, but not at the 5% level. The error correction coefficient is negative and significant, suggesting gradual adjustment toward equilibrium. The long-run coefficient on the U.S. Dollar Index is negative, consistent with the expected inverse relationship between dollar strength and crude oil prices.

Overall, ARIMA performs better as a short-run forecasting model, while ARDL provides a useful framework for interpreting the role of macro-financial variables. The findings suggest that WTI prices are difficult to forecast beyond recent price information, but the U.S. Dollar Index and the Federal Funds Rate help explain part of the dynamic behavior of crude oil prices.

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